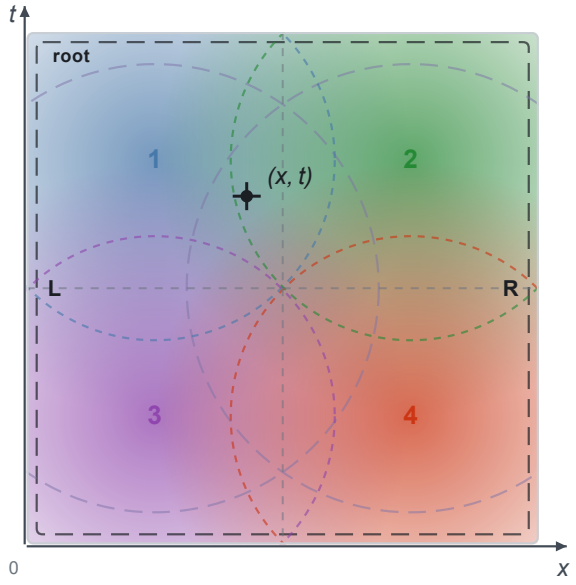
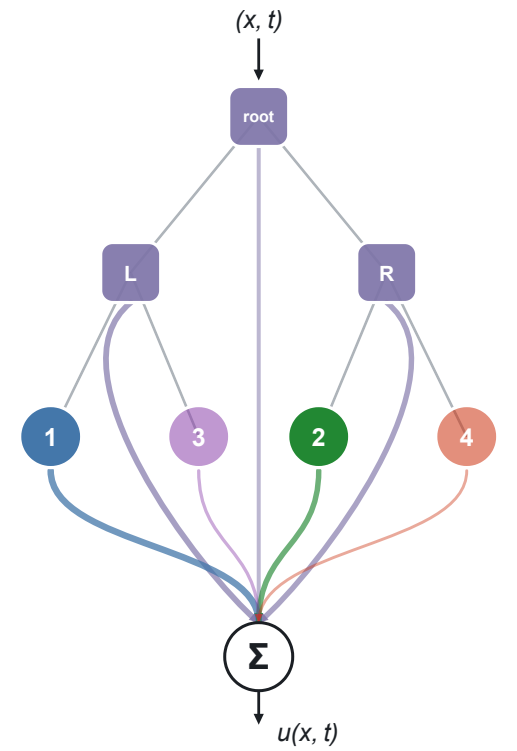
Figure 1. Inference through adaptive tree-structured mixtures of experts *a single query point (x, t) propagated through three architectures*

(A) AToE

DECOMPOSED DOMAIN



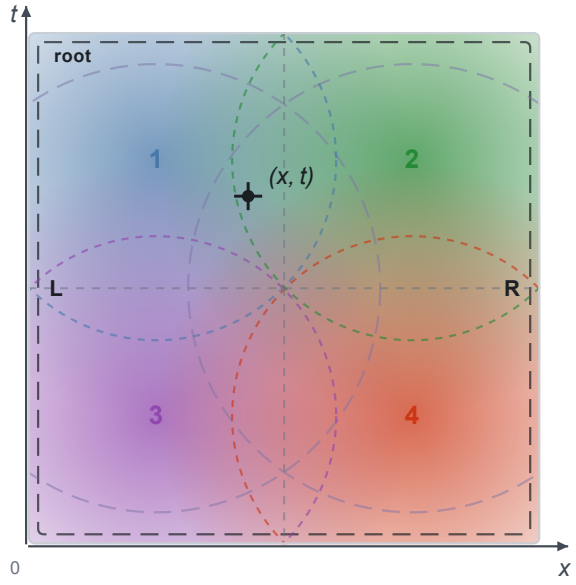
soft indicators $\tilde{\Psi}_i(x, t)$ Σ = 1.00



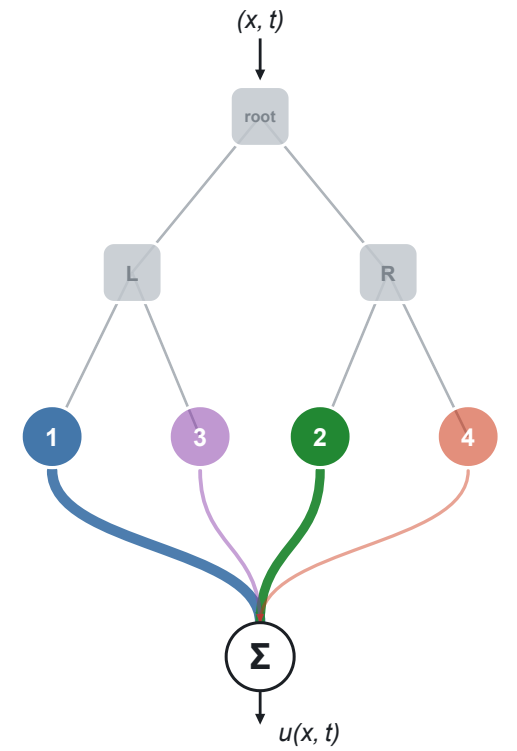
$$u(x, t) = \sum_i \tilde{\Psi}_i(x, t)\, u_i(x, t)$$

(B) AToE-Leaves

DECOMPOSED DOMAIN



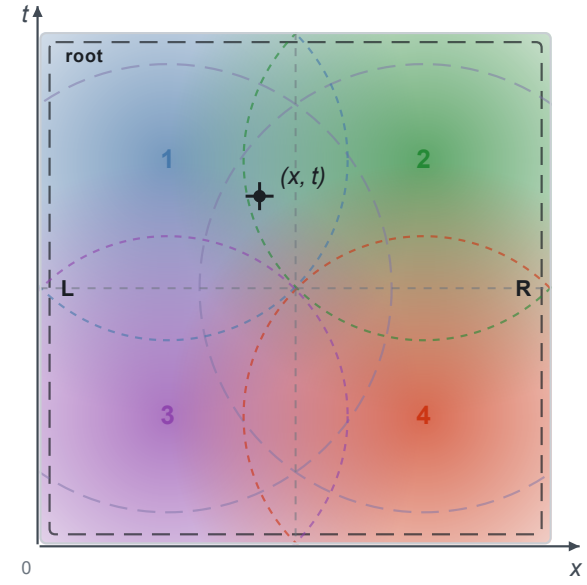
soft indicators $\tilde{\Psi}_i(x, t)$ Σ = 1.00



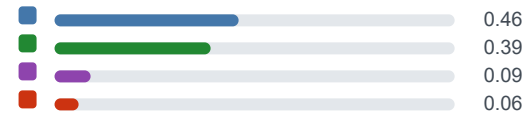
$$u(x, t) = \sum_{i \in \mathcal{L}} \tilde{\Psi}_i(x, t)\, u_i(x, t)$$

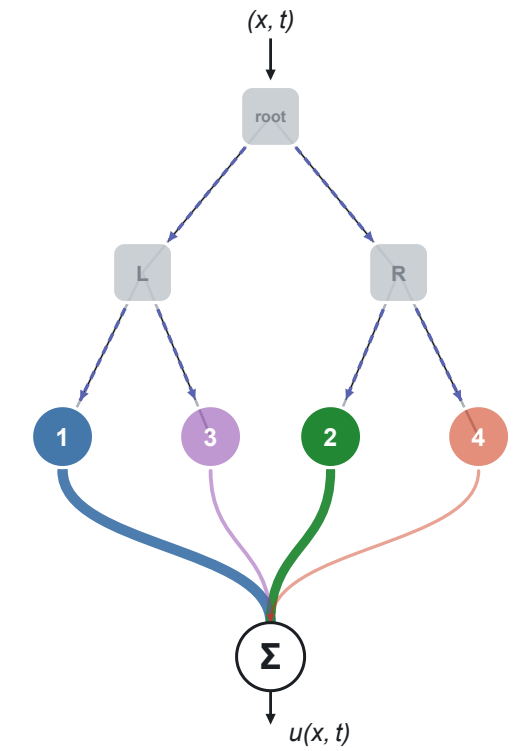
(C) ANT

DECOMPOSED DOMAIN



soft indicators $\tilde{\Psi}_i(x, t)$ Σ = 1.00





$$u(x, t) = \sum_{i \in \mathcal{L}} \tilde{\Psi}_i(x, t)\, u_i(x, t)$$

parent activations → child activations → leaf prediction

REGION ↔ NODE
- 1 = expert 1
- 2 = expert 2
- region 3
- region 4

SOFT INDICATOR
- $\tilde{\Psi}i$ peaks at the centre, fades out

SUPPORT BOUNDARY
- edge of an expert's support, where $\tilde{\Psi}i \to 0$

PARTITION OF UNITY
- overlapping supports sum to 1 everywhere

ACTIVE / INACTIVE
- coloured = active; grey □ = inactive

CONTRIBUTION & FLOW
- weight = influence
- dashed = ANT flow